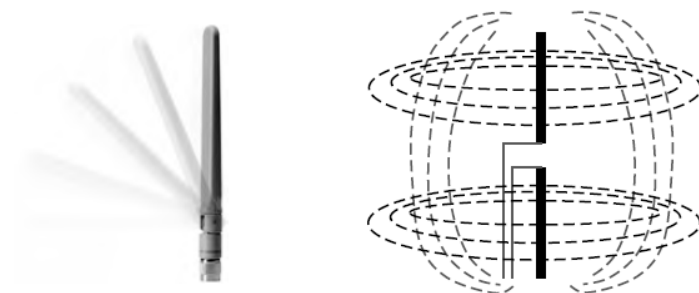


Antennas:

- o Wireless routers have different types of antennas, some routers have antennas built in.
- o Sometimes the WIFI routers will have a choice of antenna you can attach to the router.
- o Antennas come in many sizes & shapes, each with its own gain value & intended purpose.
- o There are many specific types of antennas, but three basic types are used most of time.
- o Most business class APs require external antennas to make them fully functioning units.

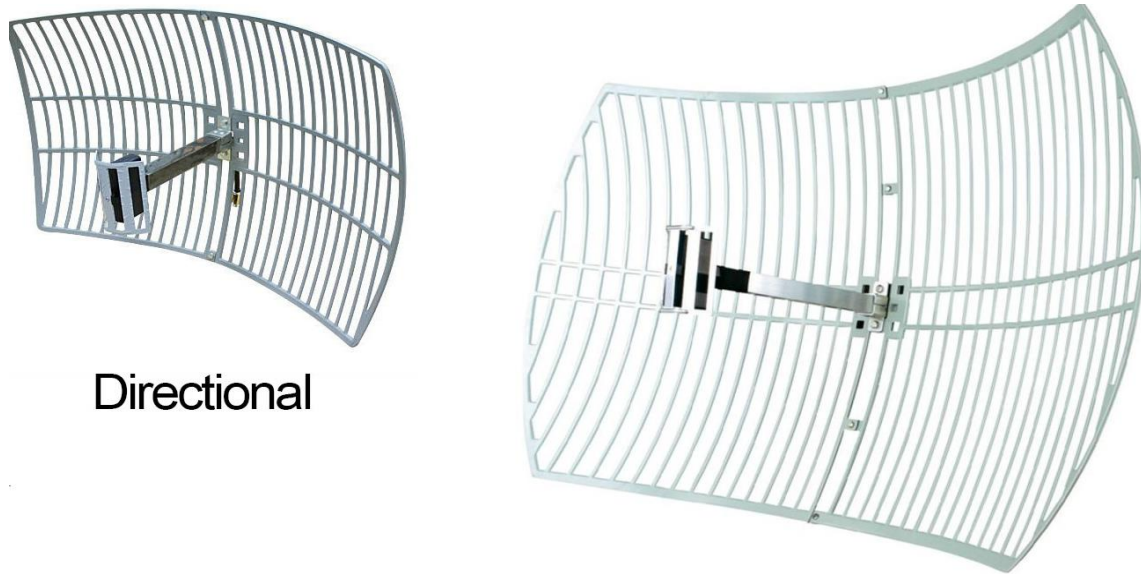
Omnidirectional Antennas:

- o An omnidirectional antenna sends a **signal out equally in all directions around it.**
- o Using omnidirectional antennas has benefit of creating connections in any direction.
- o You don't have to do as much planning to connect with multiple neighbors or buildings.
- o Omnidirectional Antenna If there is enough signal between nodes, they should connect.
- o All-direction strength of antennas comes with drawback of transmitting weaker signal.
- o Since signal is going in all directions, it spreads out & gets weaker with distance very fast.
- o If the nodes or clients, PC, Laptop or end point are far away, they may not connect well.
- o In Wireless the Omni-directional antennas are used in point-to-multipoint configurations.
- o Where they distribute the wireless signal to other computers or devices in your WLAN.
- o So, basically, a Wireless access point (AP) would use an omni-directional antenna.
- o Provide 360-degree coverage & are ideal in houses, office, conference rooms & outside.



Directional Antennas:

- o Next type of antenna is known as directional--it sends out a signal in a more focused way.
- o This Type of Antennas, Directional antennas focus the radio signal in a given direction.
- o Using directional antennas has benefit of increasing distance signal travel in one direction.
- o Power that would be sent out in all directions with omnidirectional nodes is now focused.
- o This type of Directional antennas is commonly used in point-to-point configurations.
- o This type of Directional antennas is commonly used in connecting two distant buildings.
- o Also, this Directional Antennas sometimes point-to-multipoint connecting two WLANs.
- o Few, examples of directional Wi-Fi antennas include Yagi and parabolic dish antennas.



Directional

MIMO Antenna:

- o The MIMO is wireless terms, which stand for multiple input, multiple output.
- o Basically, The MIMO is an antenna technology for wireless communications.
- o In which multiple antennas are used at both the source and the destination.
- o Antennas at each end of communications circuit are combined to minimize errors.
- o Antennas at each end of communications circuit are combined optimize data speed.
- o MIMO uses to make use of reflected signals to provide gains in channel robustness.
- o MIMO uses to make use of reflected signals to provide gains in channel throughput.
- o MIMO is uses multiple transmitters & receivers to transfer more data at the same time
- o In this Up to eight transmit and receive antennas can be used to increase throughput.

